Time: 30 minutes Max marks: 10

Instructions:
- Do not plagiarize. Do not assist your classmates in plagiarism.
- Show your full solution for the questions to get full credit.
- Attempt all questions that you can.
- In the unlikely case a question is not clear, discuss it with an invigilating TA. Please ensure that you clearly include any assumptions you make, even after clarification from the invigilator.

1. (4 points) What are points at infinity in homogeneous coordinates? Show that parallel lines intersect at a point at infinity. Make use of homogeneous coordinate based representation of lines and points and the fact that parallel lines in a 2D plane have the same slope. How would you recover the slope of the line from its corresponding point at infinity?

   Rubric:
   1 point - Defining what $\mathbf{x}_\infty$ is.
   2 points - Parallel lines intersect at $\mathbf{x}_\infty$
   1 point - Recovering the slope of the line at $\mathbf{x}_\infty$

Solution:

Part1: Defining what $\mathbf{x}_\infty$ is.

In 2D projective geometry, a point is represented in homogeneous coordinates as:

$$\mathbf{x} = (x, y, w)^T$$

If $w \neq 0$, the point corresponds to a finite point in Euclidean space:

$$(x, y) = \left(\frac{x}{w}, \frac{y}{w}\right)$$

If $w = 0$, the point is called a **point at infinity**. Such points represent directions rather than finite locations.

Thus, all points of the form:

$$(x, y, 0)^T$$

are points at infinity.

Homogeneous Representation of Lines

A line in homogeneous coordinates is represented as:

$$\mathbf{l} = (a, b, c)^T$$

and satisfies:

$$ax + by + cw = 0$$

For finite points ($w = 1$), this reduces to the familiar equation:

$$ax + by + c = 0$$

Part2: Parallel lines intersect at $\mathbf{x}_\infty$

Showing cross-product vector has a 0 for its last coordinate

Parallel Lines and Their Intersection

Consider two parallel lines in the Euclidean plane:

$$l_1 : y = mx + c_1$$

$$l_2 : y = mx + c_2$$

Rewriting in homogeneous form:

$$l_1 : mx - y + c_1 = 0 \quad \Rightarrow \quad \mathbf{l}_1 = (m, -1, c_1)^T$$

$$l_2 : mx - y + c_2 = 0 \quad \Rightarrow \quad \mathbf{l}_2 = (m, -1, c_2)^T$$

Intersection of Two Lines

The intersection point of two lines is given by the cross product:

$$\mathbf{x} = \mathbf{l}_1 \times \mathbf{l}_2$$

$$\mathbf{x} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ m & -1 & c_1 \\ m & -1 & c_2 \end{vmatrix}$$

$$= ((-1)c_2 - c_1(-1),\ c_1 m - m c_2,\ m(-1) - (-1)m)$$

$$= (c_1 - c_2,\ m(c_1 - c_2),\ 0)$$

The resulting point is:

$$\mathbf{x} = (c_1 - c_2,\ m(c_1 - c_2),\ 0)^T$$

Interpretation

Since the third coordinate is zero ($w = 0$), this is a **point at infinity**.

Normalizing (dividing by $c_1 - c_2$), we obtain:

$$\mathbf{x} \sim (1, m, 0)^T$$

Parallel lines intersect at infinity

Parallel lines have the same direction (slope m).

Their intersection is:

$$(1, m, 0)$$

This lies on the line at infinity ($w = 0$).

$$\Rightarrow \text{Parallel lines intersect at a point at infinity}$$

Part3: Recovering the slope

Recovering slope from point at infinity

Given a point at infinity:

$$\mathbf{x}_\infty = (x, y, 0)$$

From earlier:

$$\mathbf{x}_\infty \sim (1, m, 0)$$

So:

$$m = \frac{y}{x}$$

$\Rightarrow$ Slope = ratio of first two coordinates

2. (2 points) Given two sets of parallel lines in an image of a planar surface, how would we compute the homography that performs *affine rectification*, i.e., a homography that removes the projective distortion in the image?

Rubric:
0.5 for each step, 4 steps
1. Vanishing Points,
2. Vanishing Line,
3. Rectifying Homography, (0.25 for H, 0.25 for H inverse)
4. Applying Homography

Solution:

Affine Rectification using Parallel Lines

Affine rectification aims to remove projective distortion from an image of a planar surface by mapping the *line at infinity* to its canonical position:

$$\ell_\infty = (0, 0, 1)^T$$

Step 1: Compute Vanishing Points

Consider two sets of parallel lines on the plane:

- First set: lines $\mathbf{l}_1, \mathbf{l}_2$
- Second set: lines $\mathbf{l}_3, \mathbf{l}_4$

Each pair of parallel lines intersects at a vanishing point.

$$\mathbf{v}_1 = \mathbf{l}_1 \times \mathbf{l}_2$$
$$\mathbf{v}_2 = \mathbf{l}_3 \times \mathbf{l}_4$$

Step 2: Compute the Vanishing Line

The vanishing line (image of the line at infinity) is obtained by joining the two vanishing points:

$$\ell_\infty = \mathbf{v}_1 \times \mathbf{v}_2$$

Let:

$$\ell_\infty = (a, b, c)^T$$

Step 3: Construct the Rectifying Homography

We seek a homography H such that:

$$H^{-T}\ell_\infty = (0, 0, 1)^T$$

A convenient choice for H is:

$$H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ a & b & c \end{bmatrix}$$

$$H^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -a/c & -b/c & 1/c \end{bmatrix} \qquad c \neq 0$$

Step 4: Apply the Homography

Each point $\mathbf{x}$ in the image is transformed as:

$$\mathbf{x}' = H\mathbf{x}$$

Justification

Under this transformation:

$$\ell'_\infty = H^{-T}\ell_\infty = (0, 0, 1)^T$$

Thus:

- Points at infinity are restored to their canonical location
- Parallel lines in the scene become parallel in the rectified image

3. (4 points) Consider a plane Π in the 3D world. Assume that the image plane is parallel to Π. Say you have two images generated from a pure camera rotation, where the axis of rotation is the same vector as the normal to Π.

   (a) (1 point) Which type of transformation (Euclidean, Similarlity, Affine, or Projective) best describes the one between the two images? How many degrees of freedom exist in this transformation?

   (b) (2 points) Given noisy point correspondences (pairwise point matches) with outliers, write the RANSAC algorithm specifically to estimate this transformation. Include the following for full credit: (i) the transformation being estimated (ii) minimum number of correspondences to estimate this transformation (iii) the error function to be thresholded for inlier counting (iv) say if the inlier ratio is 20%, how many model hypotheses will need to be generated in RANSAC in order to estimate a reliable model?

   (c) (1 point) If we relax the assumption that the image plane is parallel to Π, how do the following change compared to the previous part? (i) degrees of freedom of the transformation (ii) minimal number of correspondences needed to estimate a model.

Solution:

(a) Rubric:
0.5 for each answer
1. Transformation,
2. DoF

Since the image plane is parallel to the plane Π, and the camera undergoes a *pure rotation* about an axis aligned with the normal to Π, the induced transformation between the two images is a **Euclidean transformation** (i.e., a pure 2D rotation).

There is:

- No translation component in the image
- No scaling or shearing

Thus, the transformation is:

$$\mathbf{x}' = R\mathbf{x}$$

where $R \in SO(2)$.

Degrees of freedom:

- A 2D rotation has **1 degree of freedom** (rotation angle)

(b) Rubric:
0.5 for each step, 4 steps
1. Transformation,
2. Min Correspondences,
3. Error function,
4. RANSAC iterations - 21 iterations
(formula correct 0.25, + if values are correctly put/correct answer 0.25, formula incorrect 0 (zero))

We estimate a **2D Euclidean transformation (rotation)** using RANSAC.

(i) Transformation:

$$\mathbf{x}' = R\mathbf{x}, \quad R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

(ii) Minimum number of correspondences:

One point correspondence is sufficient:

- A single point pair determines the rotation angle (relative to origin)

(iii) Error function:

We use the geometric reprojection error:

$$e_i = \|\mathbf{x}'_i - R\mathbf{x}_i\|_2$$

A correspondence is an inlier if:

$$e_i < \tau$$

for some threshold τ.

(iv) Number of RANSAC iterations:

Let:

- Inlier ratio: $w = 0.2$
- Sample size: $s = 1$
- Desired confidence: $p = 0.99$

Number of iterations:

$$N = \frac{\log(1-p)}{\log(1-w^s)}$$

$$N = \frac{\log(1-0.99)}{\log(1-0.2)} = \frac{\log(0.01)}{\log(0.8)} \approx \frac{-4.605}{-0.223} \approx 21$$

Thus, approximately **21 iterations** are required.

(c) Rubric:
0.5 for each answer
1. DoF,
2. Min Correspondences,

If the image plane is *not parallel* to Π, the transformation between the images becomes a **planar homography** (projective transformation). Given that nothing is said about the knowledge or calibration of the camera intrinsics, in general, the *out of plane* rotation will be a projective transformation, or a homography. If one explicitly assumes that the calibration is known, then a pure 3D rotation would have 3 degrees of freedom, and can be recovered from the homography.
(i) Degrees of freedom:

- A homography has **8 degrees of freedom**

(ii) Minimum correspondences:

- Each correspondence provides 2 constraints
- Thus, **4 point correspondences** are required

Summary:

- Parallel case $\rightarrow$ Euclidean transformation (1 DOF, 1 point)
- General case $\rightarrow$ Projective transformation (8 DOF, 4 points)